

AI Powered – Call Audit System

Case Study

Transforming Call Quality, Fraud Detection & Agent Performance with AWS AI Services

Problem Statement	Solution Overview	Application UI	Business Value Achieved
50,000–70,000+ daily calls reviewed manually Fraud cases undetected — no structured audit No real-time sentiment or dissatisfaction alerts Agent benchmarking absent — inconsistent quality	AWS AI Pipeline Call Recordings (Real-time + Batch) AWS S3 + SQS Event Triggering AWS Transcribe — Speech to Text AWS Bedrock LLM — Sentiment + Fraud	✓ Real-time & batch call ingestion ✓ Auto transcription via AWS Transcribe ✓ Bedrock LLM fraud scoring per call ✓ Sentiment trend dashboards ✓ Agent performance leaderboard ✓ AWS QuickSight compliance reports	✓ Manual reviews cut by 60–80% ✓ Fraud detection accuracy up 50–70% ✓ Real-time customer dissatisfaction alerts ✓ Consistent agent benchmarking & training ✓ Regulatory compliance tracking automated AWS Stack: S3 · SQS · Transcribe · Bedrock · Lambda · RDS · QuickSight · IAM · CloudWatch

Ganit deployed an AWS-native AI Call Audit System — automating fraud detection, sentiment analysis, and agent benchmarking across 50,000–70,000 daily calls

Outcome

Delivered a fully automated AI call auditing pipeline on AWS — ingesting real-time and batch call recordings via S3 and SQS, transcribing via AWS Transcribe, and analyzing with AWS Bedrock LLM for fraud indicators, compliance, and sentiment — surfaced through AWS QuickSight dashboards for insurance and banking operations teams.

Operational Efficiency	Cost Optimization	Scalability & Reliability	Business Growth
Reduced manual call review effort by 60% (Insurance) and 80% (Banking) — automating transcript generation, fraud scoring, and sentiment tagging across 50,000–70,000 daily calls without human reviewers.	Replaced manual QA teams with a serverless AWS pipeline (Lambda + S3 + SQS) — significantly reducing per-call review cost while scaling dynamically with call volume peaks, zero infrastructure overhead.	Built on AWS with event-driven SQS triggering, Lambda orchestration, and Amazon RDS for result storage — handling 50,000–70,000 daily calls across real-time and batch modes with IAM governance and CloudWatch monitoring.	Improved fraud detection accuracy by 50–70%, enabled proactive customer dissatisfaction management, and delivered consistent agent benchmarking — driving better training, compliance adherence, and service quality across all call centres.

High call volume, undetected fraud, and absent agent benchmarking created compliance and experience gaps for insurance operations

Problem Statement

Insurance companies process 50,000+ customer calls daily covering policy inquiries, claims, and fraud. Manual audit processes could not scale — leaving most fraud cases undetected, customer sentiment unmonitored, and agents without structured performance feedback, leading to compliance risks and increased churn.

Challenge 1

50,000+ daily calls managed without automation — fraud detection and claim approvals delayed by manual review backlog.

Challenge 2

Significant percentage of claims show inconsistencies — most fraud cases go undetected due to lack of structured call audits.

Challenge 3

Slow resolution and no real-time sentiment analysis — customer dissatisfaction escalates undetected, increasing churn.

Challenge 4

No proper agent benchmarking framework — affects training quality, service consistency, and IRDAI compliance adherence.

Ganit built a multi-source AWS AI pipeline — real-time & batch call ingestion, Transcribe speech-to-text, Bedrock LLM fraud scoring, served via QuickSight dashboards

Solution Overview

Robust & Scalable Architecture

- Real-time input via API Gateway + Lambda; batch input via ZIP file upload — both routed to S3 and SQS event triggers.
- AWS Transcribe converts call recordings to text transcripts; metadata stored in S3 for downstream analysis.
- AWS Bedrock LLM analyzes transcripts for sentiment, compliance, fraud indicators — long pauses, scripted responses, hesitation, external influence flagged.
- Results stored in Amazon RDS; Lambda orchestrates end-to-end with CloudWatch monitoring and IAM governance.

Application UI

- AWS QuickSight dashboards for real-time fraud monitoring, agent performance evaluation, and compliance tracking.
- Dual input modes: API Gateway (real-time streaming) and batch ZIP uploads for historical call processing.
- Row-level security and role-based access via IAM — separate views for compliance, QA, and management teams.

Data Architecture

